

Calibration Certificate

Sample: Zenith Polymer® Diffuse Reflectance Standard – 99%R
Type: SG3051
Serial-No.: 23RW0349
Report-No.: 23041302
PO-No.: PO23040
Customer: Pro-Lite Technology France SAS
10 Avenue Roger Lapébie, Bâtiment B, 1er étage
33140 Villenave d'Ornon
France

Test and Test Method

The 8°/hemispherical spectral reflectance was measured for the sample listed above. The calibration was performed with a PerkinElmer Lambda 950 UV-VIS-NIR spectrometer (serial number 950L1211082) equipped with a 150 mm integrating sphere. The reflectance was determined by using a standard from SphereOptics GmbH, part number SG 3051, which has been calibrated by the National Metrology Institute of the Federal Republic of Germany (PTB).

Measuring Equipment

Reference:	Lambda 950 Spectrometer	PTB Reference Standard
Reference No.:	WO-02222533	PTB 44135/22
Last maintenance:	February 27, 2023	November 16, 2022
Next maintenance:	February 2024	November 2023

Total Calibration Uncertainty

The total uncertainty for this measurement is:

Wavelength Range	Total Uncertainty (95 % confidence)
250-300 nm	1.11 %
>300-1750 nm	0.42 %
>1750-2200 nm	0.57 %
>2200-2400 nm	1.71 %
>2400-2450 nm	2.51 %

Laboratory Environment

Temperature: 23 °C ± 2 °C
Humidity: 45 % ± 15 %

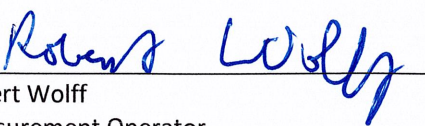
Calibration Documentary

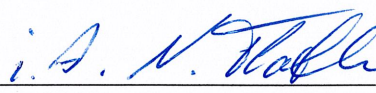
This calibration certificate includes the printed spectral reflectance curve of the sample from 250-2450 nm reported in 50 nm intervals, together with the electronic calibration data delivered on a portable data storage device.

Calibration Interval

If the product is handled with care, recalibration is recommended one year after the date of this certificate, at the discretion of the user.

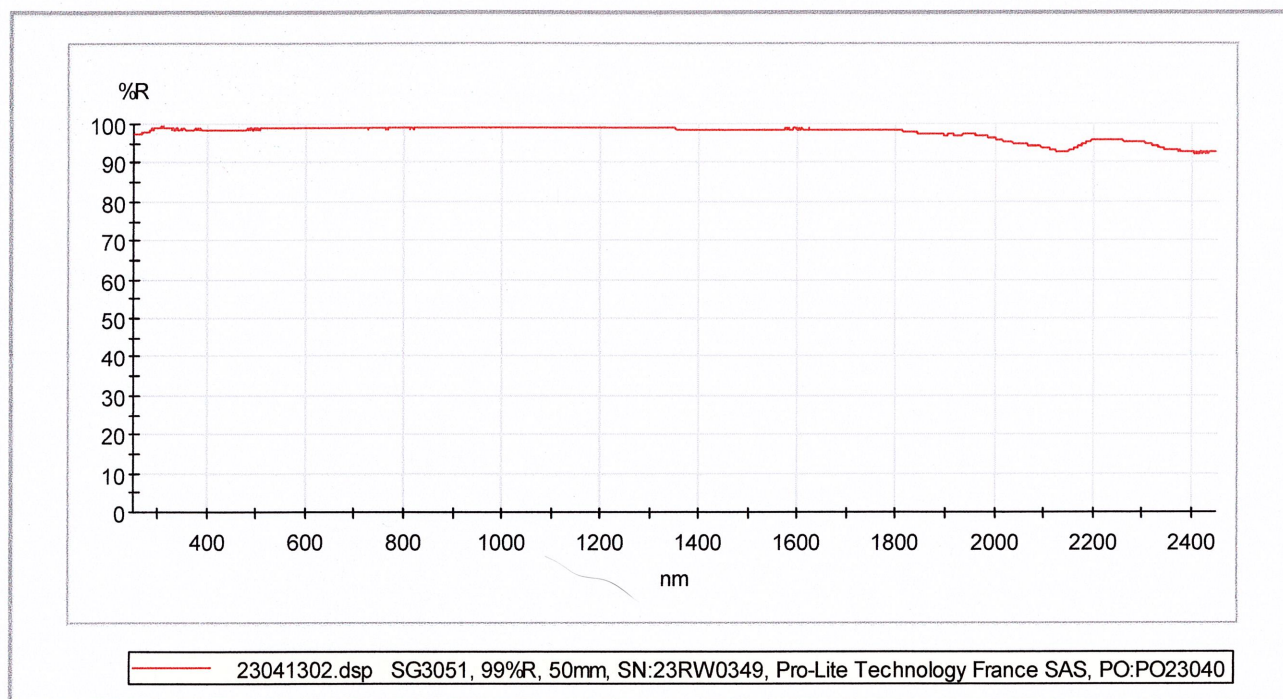
Date: April 13, 2023


Robert Wolff
Measurement Operator

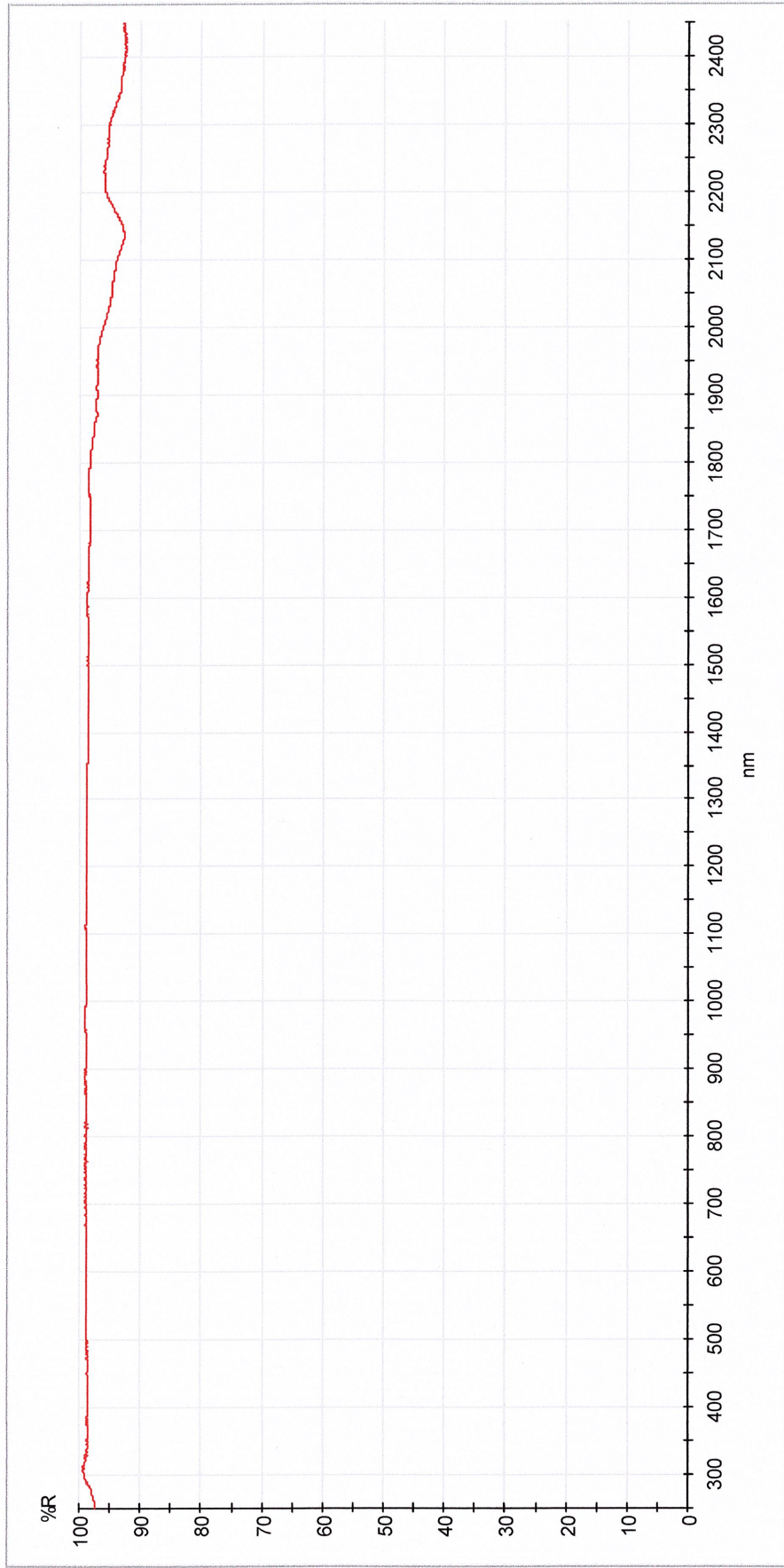

Daniel Schwefel
Head of Laboratory

Diffuse Reflectance Measurement

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Abscissa [nm]	Ordinate [%R]	Abscissa [nm]	Ordinate [%R]	Abscissa [nm]	Ordinate [%R]
250	97,333	1.000	98,921	1.750	98,387
300	99,045	1.050	98,936	1.800	98,342
350	98,713	1.100	98,936	1.850	97,646
400	98,565	1.150	98,941	1.900	97,096
450	98,685	1.200	98,937	1.950	97,216
500	98,776	1.250	98,911	2.000	96,177
550	98,800	1.300	98,888	2.050	94,712
600	98,811	1.350	98,765	2.100	93,944
650	98,938	1.400	98,536	2.150	93,007
700	98,942	1.450	98,637	2.200	95,777
750	98,782	1.500	98,694	2.250	95,759
800	98,786	1.550	98,668	2.300	95,244
850	98,889	1.600	98,706	2.350	93,316
900	98,978	1.650	98,539	2.400	92,667
950	98,915	1.700	98,270	2.450	92,620



23041302.dsp SG3051, 99%R, 50mm, SN:23RW0349, Pro-Lite Technology France SAS, PO:PO23040

Slit: 4 nm

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